Notes about GRAVITY+ metrology demodulation

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1 The metrology table

The metrology data is stored as a FITS Table in the `METROLOGY` HDU (10th) as in table 1. The voltages `VOLT` are composed of 80 columns:

- 2 directions (x and y) per diodes
- 4 diodes per telescope (one on each spiders)
- 1 fiber coupler diode (labeled `FC`) per telescope
- 4 telescopes
- 2 sides: `FT` and `SC`

The signals (the table column) are sample at 500 Hz leading to a very large number of rows.

2 The modulation model

When the pupil modulation is on (`ESO INS PMC1 MODULATE` keyword is `true`), the metrology signal is modulated at a frequency of $f = 1\,\text{Hz}$. This modulation does not affect the fiber coupler diode.

Name	Size	Type	TFORM
TIME		Int32	1J
VOLT	(80,)	Float32	80E
POWER_LASER		Float32	1E
LAMBDA_LASER		Float32	1E

Table 1: Metrology table

For each diode in {D1, D2, D3, D4}, we define the complex measurement $\boldsymbol{v} \in \mathbb{C}^N$ with $v_i = x_i + \jmath y_i\,, \quad \forall i \in [\![1, N]\!]$ and N is the number of row of the metrology. The modulation as a function of time is:

$$\boldsymbol{v} = \exp\left(\jmath\, b \sin\left(\boldsymbol{\omega} + \phi\right)\right)\,. \tag{1}$$

where $\omega = 2\,\pi$ is the modulation pulsation, the amplitude b and the phase ϕ parameters of the modulation has to be estimated to demodulate the metrology signal.

3 The overall model

For a diode the measured modulated voltage data $\boldsymbol{d}$ can be modeled by:

$$\boldsymbol{d} = (c + \boldsymbol{s} \times \exp\left(\jmath\, b \sin(\boldsymbol{\omega} + \phi)\right)) \times \exp\left(\jmath\, \Phi_{\mathrm{FC}}\right) + \boldsymbol{e}\,, \tag{2}$$

where $\times$ is the element-wise multiplication, $\boldsymbol{s}$ is the sought-after demodulated metrology signal, Φ_{FC} is the phase of the fiber coupler measurement accounting for FDDL movements. $c = x_0 + \jmath\, y_0$ is the center of the pupil and $\boldsymbol{e}$ is a vector representing the errors on the measurement that are supposed Gaussian centered, independent and identically distributed.

4 Estimating the modulation parameters

To estimate the modulation parameters we need to disentangle it from the signal $\boldsymbol{s}$. This is done using the expansion:

$$\boldsymbol{s} = a + \boldsymbol{\delta v} \tag{3}$$

$|a|$ and $\arg(a)$ being the mean amplitude and phase respectively of the metrology signal. $\boldsymbol{\delta v}$ is centered ($\mathrm{E}(\delta v) = 0$) and supposed to be Gaussian independant and identically distributed.

Under this assumption, the modulation parameters (b, ϕ), the center c and the mean of the metrology signal a can be estimated using least square:

$$(c^+, a^+, b^+, \phi^+) = \underset{c,a,b,\phi}{\arg\min}\, \|\boldsymbol{d} - (c + a \exp\left(\jmath\, b \sin(\boldsymbol{\omega} + \phi)\right)) \times \exp\left(\jmath\, \Phi_{\mathrm{FC}}\right)\|_2^2 \tag{4}$$

that is equivalent with :

$$(c^+, a^+, b^+, \phi^+) = \underset{c,a,b,\phi}{\arg\min}\, \|\boldsymbol{r} - c - a \exp\left(\jmath\, b \sin\left(\boldsymbol{\omega} + \phi\right)\right)\|_2^2 \tag{5}$$

with $\boldsymbol{r} = \boldsymbol{d} \times \exp\left(-\jmath\, \Phi_{\mathrm{FC}}\right)$.

4.1 Linear estimates

From the equation 5, we can see that the parameters c and a depends linearly of the $\boldsymbol{d}$ and the modulation phasor $\boldsymbol{m} = \exp\left(\jmath\, b \sin\left(\boldsymbol{\omega} + \phi\right)\right)$. For a given values of b and ϕ, the optimal values c^+ and a^+ have a closed-form solution that is:

$$\begin{bmatrix} c^+ \\ a^+ \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} \sum_i d_i \\ \boldsymbol{m}^{\mathrm{T}} \boldsymbol{d} \end{bmatrix} \tag{6}$$

where $\boldsymbol{m}^{\mathrm{T}}$ is the conjugate transpose of $\boldsymbol{m}$ and $\mathbf{A}$ is the 2×2 matrix:

$$\mathbf{A} = \begin{bmatrix} N & \sum_i m_i \\ \sum_i m_i^* & N \end{bmatrix}, \tag{7}$$

where m_i^* is the complex conjugate of m_i and N the number of measurements. Its inverse is:

$$\mathbf{A}^{-1} \frac{1}{N^2 \left(1 - \left|\sum_i d_i\right|^2\right)} \begin{bmatrix} N & -\sum_i m_i \\ -\sum_i m_i^* & N \end{bmatrix} \tag{8}$$

For a couple (b, ϕ) the optimal $c^+(b, \phi)$ and $a^+(b, \phi)$ is given by:

$$c^+(b, \phi) = \frac{N \sum_i d_i - \boldsymbol{m}^{\mathrm{T}} \boldsymbol{d} \sum_i m_i}{N^2 \left(1 - \left|\sum_i d_i\right|^2\right)} \tag{9}$$

$$a^+(b, \phi) = \frac{-N \left(\sum_i d_i\right) \left(\sum_i m_i^*\right) + \boldsymbol{m}^{\mathrm{T}} \boldsymbol{d}}{N^2 \left(1 - \left|\sum_i d_i\right|^2\right)} \tag{10}$$

4.2 Non-linear estimates

The modulation parameters estimation amounts to estimate only b and ϕ, optimizing the function:

$$f(b, \phi) = \left\| \boldsymbol{r} - c^+(b, \phi) - a^+(b, \phi) \exp\left(\jmath\, b \sin\left(\boldsymbol{\omega} + \phi\right)\right) \right\|_2^2 \tag{11}$$

This function is non-linear and non-convex as it can be seen on figure 1. The two main optimum are equivalent as $f(b, \phi) f(-b, \phi + \pi)$ An initialization with b sufficiently small seems to ensure to end in the global optimum. Note that the case $b = 0$ is singular as $f(b, \phi)$ is equal to the variance of $\boldsymbol{r}$ in this case whatever is ϕ, a and c.

In the code this function is minimizes by the mean of the derivative free NEWOA method of powel that seems faster than VMLMB (with derivative) and the Simplex method.

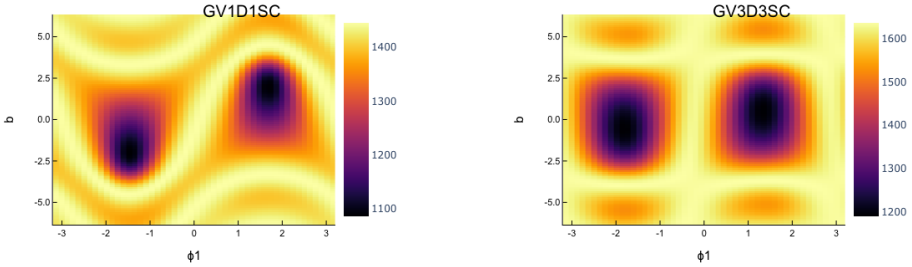

Figure 1: $f(b, \phi)$ for two different diodes

Name		keyword
center	$\mathrm{Re}(c^+)$	`DEMODULATION CENTER X0`
center	$\mathrm{Im}(c^+)$	`DEMODULATION CENTER Y0`
metrology amplitude	$\lvert a^+ \rvert$	`DEMODULATION AMPLITUDE ABS`
metrology phase	$\arg(a^+)$	`DEMODULATION AMPLITUDE ARG`
modulation amplitude	b^+	`DEMODULATION SIN AMPLITUDE`
modulation phase	ϕ^+	`DEMODULATION SIN PHASE`

Table 2: Metrology table

5 Demodulation

Once the modulation parameters are estimated the demodulated signal is given by:

$$\boldsymbol{s} = \left(\boldsymbol{r} - c^+\right) \times \exp\left(\jmath\, b^+ \sin(\boldsymbol{\omega} + \phi_1^+)\right) \times \exp\left(\jmath\, \Phi_{\mathrm{FC}}\right) , \tag{12}$$

All the $6 \times 4 \times 4 \times 2$ parameters are stored in keywords of the METROLOGY table header as shown on the table 2. These keywords are suffixed with the side, the telescope number and the diode (*e.g.* `DEMODULATION CENTER X0 SC T4 D1`).